Report R16 — the rules that sit under the $ab \geq 2$ hyperbola, and why

Krylov sector analysis of quantum cellular automata

Tier 1 analysis

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Abstract

A handful of markers in the left panel of R9's Fig. F1 lie below the exclusion curve $ab = 2$, which the partition of the computational basis forbids. This report identifies them and finds **two unrelated causes**. The larger effect is a *rendering artefact*: F1 nudges the three families sideways by ± 0.019 in a so that none hides under another, and because $b = 2/a$ rises as a falls, the 40 V-free rules sitting exactly *on* the curve at $(1, 2)$ are drawn where the curve is 2.0387 — under it. They are the biggest marker in the panel. The smaller effect is real arithmetic: **seven** rules — W_6, W_{14}, W_{20}, W_{74}, W_{84}, W_{88}, W_{229} — had a fitted product $ab = 1.919$–1.937. The answer to "is it simply a fitting problem?" is *yes, but the fix is a theorem rather than a better fit.* Their sector counts are exact quasi-linear staircases, $\lceil N/2 \rceil + 1$ and $\lfloor (N + 4)/3 \rfloor$, verified at every $N \leq 22$; so $a = 1$, pigeonhole gives $D_{\max} \geq 2^N/n_{\mathrm{wcc}}$ and hence $b \geq 2$, while $D_{\max} \leq 2^N$ gives $b \leq 2$. The pair $(1, 1.92)$ is not a violated theorem but an *internally inconsistent descriptor*. No better estimator rescues the fit on this window: the certified control W_{134}, whose $D_{\max}$ *is* the central binomial and whose base is 2 by derivation, is itself fitted at 1.9244 and has a rolling base of 1.904–1.938, squarely inside the band spanned by the seven. We extended the sweep to $N = 22$, installed derived $(a, b) = (1, 2)$ overrides with a refitted prefactor power, and the sector map now reports 204 rules on the curve, 50 above, 2 irregular, and **zero** below.

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1 The question

R9 §"The hyperbola" derives the only inequality in this project that constrains the sector map from outside the data. The weak components partition the 2^N basis states, so the largest one cannot be smaller than the average:

$$n_{\text{wcc}}(N)\, D_{\max}(N) \;\geq\; 2^N, \tag{1}$$

and writing $n_{\text{wcc}} \sim a^N$, $D_{\max} \sim b^N$ and taking N-th roots,

$$a\, b \;\geq\; 2. \tag{2}$$

Equality holds when the sectors are comparable in size. The curve is drawn in the left panel of R9's F1 and it excludes the region below it. Some markers are down there. This report asks which, and why.

The short answer is that "some markers are down there" conflates two things that have nothing to do with each other, and separating them is most of the work.

2 Cause one: a rendering artefact, and it is the biggest marker

F1 stacks 256 rules into few cells — 77% of the sector map is a single point — so `sector_figure` offsets the three families horizontally to stop one hiding under another:

$$\Delta a = \begin{cases} +0.019 & \text{V+reset} \\ 0.000 & \text{unitary} \\ -0.019 & \text{V-free.} \end{cases}$$

This is documented in F1's caption as a caveat about reading absolute bases off the axis. At the top-left corner it is more than that. The exclusion curve is *steep* there: $\mathrm{d}b/\mathrm{d}a = -2/a^2 = -2$ at $a = 1$, so a leftward nudge of 0.019 raises the curve by 0.0387 while leaving the marker at $b = 2$. Every V-free rule at exactly $(1, 2)$ — one sector of size 2^N, on the curve to machine precision — is therefore *drawn* at $(0.981, 2.000)$ with the curve passing above it at 2.0387.

There are 40 such rules after the correction of §3 (35 before it; five of the seven are V-free and join this cell). They are the single largest marker in the panel, and they account for more visual "violation" than the real effect does. The verdict table has always reported them correctly — their product is 2.000000 — so nothing downstream was ever wrong; only the picture was misleading. F1's caption and figure header now say so.

3 Cause two: seven rules whose fitted product really was below 2

3.1 Who they are

Lifting the overrides described below and re-running the 256-rule fit inside the uniform window $N \leq 16$ gives exactly seven rules with $ab < 2$:

group	rules	tuples	family
A	W_6, W_{14}, W_{20}, W_{84}	`IVDD`, `IEDD`, `IDVD`, `IDED`	$6, 20$ mixed; $14, 84$ V-free
B	W_{74}, W_{88}, W_{229}	`DEID`, `DIED`, `EDIE`	all V-free

The grouping is not cosmetic: within each group the rules share an *identical* n_{wcc} series and an identical $D_{\max}$ series at every N computed, so there are four distinct series among the seven, not seven. Group A is fitted at $b = 1.922559$ and group B at 1.932218 (W_{74}), 1.936760 (W_{88}), 1.918755 (W_{229}); all seven have $a = 1$.

3.2 The sector counts are exact, not fitted

Proposition 1. *For every N on record ($6 \le N \le 22$, computed at open boundaries),*

$$n_{\mathrm{wcc}}(N) = \left\lceil \frac{N}{2} \right\rceil + 1 \quad \textit{(group A)}, \qquad n_{\mathrm{wcc}}(N) = \left\lfloor \frac{N+4}{3} \right\rfloor \quad \textit{(group B)}.$$

This is checked exhaustively rather than fitted, and it is also recovered independently by a residue-class solver (`below_curve.quasilinear_formula`) that looks for an affine form in N on each class modulo $m \le 4$ using exact rational arithmetic; it returns period 2 with common slope 1/2 for group A and period 3 with common slope 1/3 for group B. So $n_{\mathrm{wcc}} = \Theta(N)$ and, unambiguously, $a = 1$.

3.3 The squeeze

Proposition 2 ($b = 2$ exactly)**.** *If $n_{\mathrm{wcc}}(N) = \Theta(N)$ then*

$$b \;=\; \lim_{N\to\infty} D_{\max}(N)^{1/N} \;=\; 2.$$

Proof. Pigeonhole on the partition gives $D_{\max} \ge 2^N/n_{\mathrm{wcc}} \ge 2^N/(cN)$ for some constant c, hence $b \ge \lim(2^N/cN)^{1/N} = 2$. A single sector cannot exceed the whole basis, so $D_{\max} \le 2^N$ and $b \le 2$. $\square$

Both halves are elementary; the point is that *once $a = 1$ is established, b is not a free parameter at all.* The fitted pair $(1, 1.92)$ does not violate (2) so much as contradict itself: the same partition argument that draws the curve fixes the second coordinate given the first.

It is worth stating plainly why the finite-N version of the bound never bites. Equation (1) holds comfortably — the smallest value of $n_{\mathrm{wcc}}D_{\max}/2^N$ across the seven is 2.19 (group A), 1.31 (W_{74}, W_{88}) and 1.59 (W_{229}), and across all 256 rules the tightest case anywhere is W_0 at $N = 6$ with ratio exactly 1.0000. But the induced bound on the base, $(2^N/n_{\mathrm{wcc}})^{1/N}$, converges to 2 at the glacial rate $(cN)^{-1/N}$: at $N = 22$ it is still only 1.79–1.82. The theorem is exact and the finite-N shadow of it is useless. That is the whole difficulty.

4 Why the fitter missed it

4.1 The branch that was supposed to catch this

`sectors.series_descriptor` contains a branch, `_volume_fraction_base`, whose entire job is this case: it asks whether the residual $D_{\max}/2^N$ is better explained by a power of N (in which case the base is exactly 2 and α comes from that fit) or by an exponential in N (in which case the base is genuinely below 2). It requires the power model to win on the last six points *and* on the full series, and refuses if the series falls back down or if $|\alpha| > 8$.

It refused here, and it refused by a hair. Table 4.1 reports the relative margin of the tail comparison:

rule	tuple	n_{wcc} exactly	fitted b	fitted ab	tail margin	power wins full	tail secant	over N	α at $b = 2$
W_6	`IVDD`	$\lceil N/2 \rceil + 1$	1.9226	1.9226	-1.3%	yes	1.9671	14…22	-0.413
W_{14}	`IEDD`	$\lceil N/2 \rceil + 1$	1.9226	1.9226	-1.3%	yes	1.9671	14…22	-0.413
W_{20}	`IDVD`	$\lceil N/2 \rceil + 1$	1.9226	1.9226	-1.3%	yes	1.9671	14…22	-0.413
W_{74}	`DEID`	$\lfloor (N+4)/3 \rfloor$	1.9322	1.9322	-0.2%	yes	1.9562	15…21	-0.295
W_{84}	`IDED`	$\lceil N/2 \rceil + 1$	1.9226	1.9226	-1.3%	yes	1.9671	14…22	-0.413
W_{88}	`DIED`	$\lfloor (N+4)/3 \rfloor$	1.9368	1.9368	-13.0%	no	1.9562	15…21	-0.326
W_{229}	`EDIE`	$\lfloor (N+4)/3 \rfloor$	1.9188	1.9188	-1.0%	yes	2.0052	15…21	-0.437

Group A loses the six-point tail test by 1.3% in residual sum — 0.00682 against 0.00673 — while *winning* the full-series test by 18.9%. W_{74} loses the tail by 0.2% and W_{229} by 1.0%. Only W_{88} loses both, and it loses the full test by 14.7%. Five of the seven would have been caught by a rule that asked the full series alone.

The reason the two windows disagree is a genuine near-degeneracy, not a bug:

$$c \cdot 2^N N^{-0.41} \quad \text{and} \quad 1.9226^N \tag{3}$$

agree to better than 2% over $N = 6 \dots 16$, because $\ln N$ is very nearly affine in N over a window that short. Loosening the tail test until it accepted group A would be tuning a threshold to a known answer, and it would then have to be defended against every other rule in the sweep.

4.2 The certified control settles that no estimator would do better

The decisive check is a rule whose base is known by derivation rather than by fit. W_{134}'s $D_{\max}$ is exactly rule 150's central binomials $924, 1716, 3003, 6435, 12870, 24310, \dots$, i.e. $\binom{N}{\lfloor N/2 \rfloor} \sim 2^N/\sqrt{\pi N/2}$, so its base is 2 with $\alpha = -1/2$ and there is nothing to discover. Fitted with a pure exponential over the same window it returns 1.9244, and its rolling six-point base runs 1.904–1.938 — *inside* the range 1.858–2.039 that the seven span over the same window. There is no residual signal in the fit that separates the two cases.

control	what it is	$N_{\max}$	widest secant	tail secant	secant > 2
W_{134}	central binomials, $b = 2$ by derivation	17	1.9278	1.9856	no
W_{28}	a genuine sub-2 base	17	1.7056	1.7321	no

The complementary control is W_{28}, whose base really is below 2: it sits flat at 1.68–1.73 and never approaches the band. So the estimator is not broken — it separates a real sub-2 base from base 2 perfectly well — it simply cannot separate 2-with-a-power-law-prefactor from 1.92 on $N \leq 16$. Deriving b is not a workaround for a weak fit; it is the only thing that can settle the question at these sizes.

5 The extended sweep to $N = 22$

The seven were re-run to $N = 22$ (background sweep, `wcc_sweep --lax`), giving a lever arm two-thirds longer than the uniform window. Three things came out of it, and one of them cuts against the tidiest version of the story.

The staircases survive. Both closed forms hold at every new N. This is the load-bearing fact — everything else rests on $a = 1$ — and it is now checked over 17 system sizes rather than 11.

Group A's fitted base climbs like the control's. The pure-exponential base fitted on $N = 6 \dots M$ rises monotonically with M for W_6, W_{14}, W_{20}, W_{84}: 1.8951 at $M = 10$ to 1.9345 at $M = 22$, tracking the analytically continued central binomial ($1.8867 \to 1.9291$) almost step for step, while a synthetic series with a genuine base 1.9226 is flat at 1.9226 by construction. That is the signature of a power-law prefactor on base 2.

Group B's does not, and we should not pretend otherwise. For W_{74}, W_{88} and W_{229} the same estimator wanders or drifts *down*: W_{74} goes $1.9884 \to 1.9313$. The cause is visible in the data — their n_{wcc} staircase has period 3 and $D_{\max}$ inherits a strong modulation from it (W_{74}'s residual alternates by nearly 50% between odd and even N) — and a base fitted across residue classes absorbs part of the modulation. The climbing argument is therefore corroboration for group A only. Panel (c) of Fig. 1 is titled accordingly.

A model-free refutation, for one rule. Restricting to a residue class removes the modulation and enables a diagnostic that needs no model at all. For a pure exponential $c\,b^N$, *every* secant

$$\left(\frac{D_{\max}(N_2)}{D_{\max}(N_1)}\right)^{1/(N_2-N_1)}, \qquad N_1 \equiv N_2 \pmod{m},$$

equals b exactly. All seven exceed their own fitted base over the tail. And W_{229}'s odd branch *stops decaying* relative to 2^N at $N=11$ and then rises — $D_{\max}/2^N = 0.3867, 0.3965, 0.4219, 0.4341, 0.4355, 0.428$ at $N = 11, 13, 15, 17, 19, 21$ — giving a secant of **2.0052** over $N = 15 \ldots 21$. A base below 2 forces $D_{\max}/2^N = c(b/2)^N$ to decay geometrically forever, so this single branch refutes the fitted model for W_{229} from the data alone, with no asymptotics and no theorem. The same diagnostic returns 1.7321 for W_{28}, so it is not firing on noise.

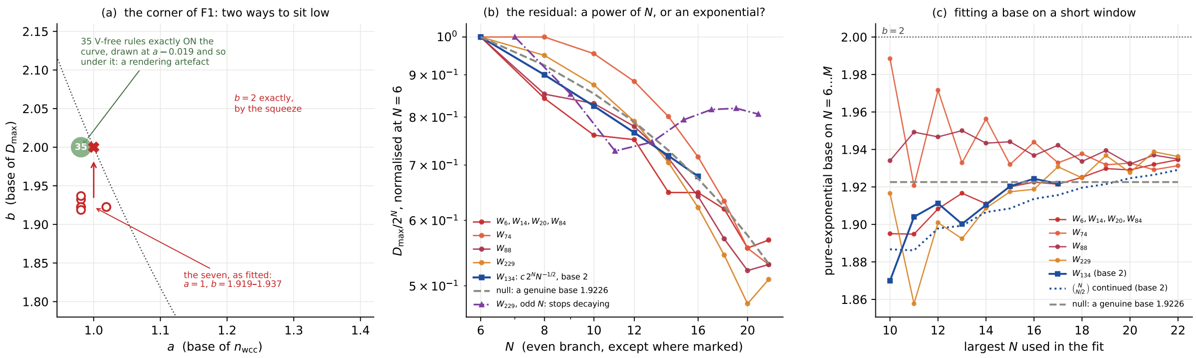

Figure 1: **Two ways to sit under the curve.** (a) The corner of R9's F1. The green cell is 40 V-free rules at exactly $(1,2)$ — on the curve — drawn at $a = 0.981$ where the curve is 2.0387, and so *below* it: a rendering artefact (§2). The open red markers are the seven as fitted; the arrow is the correction to the derived $(1,2)$. (b) The residual $D_{\max}/2^N$ on log–log axes, normalised at $N=6$, even branch: a power-law prefactor is a straight line, a genuine base bends. The certified control W_{134} (blue squares, base 2 by derivation) and the null (grey dashed, a genuine base 1.9226) bracket the question, and over this window they barely separate. The purple branch is W_{229} at *odd* N, which stops decaying — the model-free refutation of §5. (c) The pure-exponential base fitted on $N = 6 \ldots M$ against M. The null is flat by construction; the continued central binomial climbs; group A climbs with it; group B wanders, because its period-3 modulation contaminates a base fitted across residues.

6 The correction

Seven `ANALYTIC` entries were installed in `scaling/sectors.py`, keyed `(rule, "obc0")`:

$$n_{\mathrm{wcc}}:\ (a,\alpha) = (1,1), \qquad D_{\max}:\ b = 2,\ \alpha = \texttt{None}.$$

The `None` is deliberate and required a small extension to `series_descriptor`. An analytic entry may derive the *base* without deriving the prefactor power that goes with it; writing a fitted number into a table headed “derived rather than fitted” would be worse than refitting it openly. When α is `None` the power is now refitted with the derived base held fixed, via the existing `_alpha_at_base`, and the descriptor records `alpha_source = "refit at the derived base"`. The results are $\alpha = -0.413$ (group A), -0.295 (W_{74}), -0.326 (W_{88}), -0.437 (W_{229}) — all comfortably inside the $|\alpha| \le 8$ guard and all of the size a prefactor power should be.

Effect on the sector map.

verdict	before	after
on the curve, $ab = 2$	197	204
above, $ab > 2$	50	50
sub-exponential factor, test degenerate	7	0
irregular series, no growth law	2	2
rules with $ab < 2$	**7**	**0**

The anchors are untouched: W_{204}, W_{51} and W_{150} remain at $ab = 2.000000$ and the six room-packing rules remain at $\varphi{\cdot}4^{1/5} = 2.1350$. The full validation pass reports `hyperbola_violations: 0` and `sum_rule_max_abs_residual: 0` over 2926 rule–N units.

Corroborating structure. The hyperbola is tight when the sectors are comparable in size, which is what these rules look like. W_{14} at $N = 16$ has nine sectors of sizes 23234, 18250, 13456, 5666, 3830, 550, 5 summing to 2^{16}: the top five are within a factor of six of each other and carry 98.3% of the basis. That is the equality condition of (2) — comparable sectors, a bounded number of them — not the profile of a rule with a smaller base.

7 Answering the question as asked

1. **Which rules?** Seven have a genuinely sub-2 fitted product: W_6, W_{14}, W_{20}, W_{74}, W_{84}, W_{88}, W_{229}, in two groups of four and three sharing identical series. Separately, 40 V-free rules that are exactly on the curve are *drawn* below it by F1's family offset; that marker is larger than the real effect and is the first thing a reader sees.

2. **Is it simply a fitting problem?** Yes — in the precise sense that no sector count and no sector size is wrong, the finite-N bound (1) holds at every N for all 256 rules, and only the summary pair (a, b) was wrong. But "simply" would be misleading if it suggested a better fit is the remedy. The certified control shows that on $N \leq 16$ (indeed on $N \leq 22$) no pure-exponential estimator can separate base 2 with an $N^{-0.4}$ prefactor from a genuine base 1.92. The remedy is the theorem: $a = 1$ is exact from the staircase, and $a = 1$ forces $b = 2$.

3. **What changed?** Seven derived overrides, one new refit path for analytic entries that fix a base but not a power, a corrected note in F1's caption and header, and a sector map with nothing below the curve.

8 Reproduction

```
# the analysis, its figure and its two tables
python -m qca_fragmentation.scaling.below_curve --bc obc0 --rebuild

# the sector map and R9's figures/tables, which consume the overrides
python -m qca_fragmentation.scaling.sector_figure --bc obc0 --rebuild
python -m qca_fragmentation.scaling.r9_tables     --bc obc0

# the extension sweep behind Section 5 (already on disk to N=22)
python -m qca_fragmentation.wcc_sweep --rules 6,14,20,74,84,88,229 \
       --bc obc0 --n-min 17 --n-max 22 --lax

# the regression tests
python -m pytest qca_fragmentation/tests/test_below_curve.py -q
```

scaling/below_curve.py is written so that it can still explain what it corrected after the correction is in place: sectors.R16_OVERRIDES names exactly the entries R16 added, and the context manager without_r16_override lifts them so the pre-correction fit is recomputed from the series rather than hard-coded into the prose. Every number in this report comes out of analytics/below_curve_obc0.json.

The regression suite pins the parts that could rot silently: that R16_OVERRIDES still matches the set of rules actually below the curve, that lifting the override is reversible, that both closed forms still hold at every N, that below_curve.FAM_DX still equals sector_figure._FAM_DX (otherwise §2 would be explaining a figure that no longer exists), that group A still climbs and group B still does not, and that the α overrides parse under matplotlib's mathtext as well as under LaTeX — a test which caught a real defect, \ge being valid LaTeX but not valid mathtext.